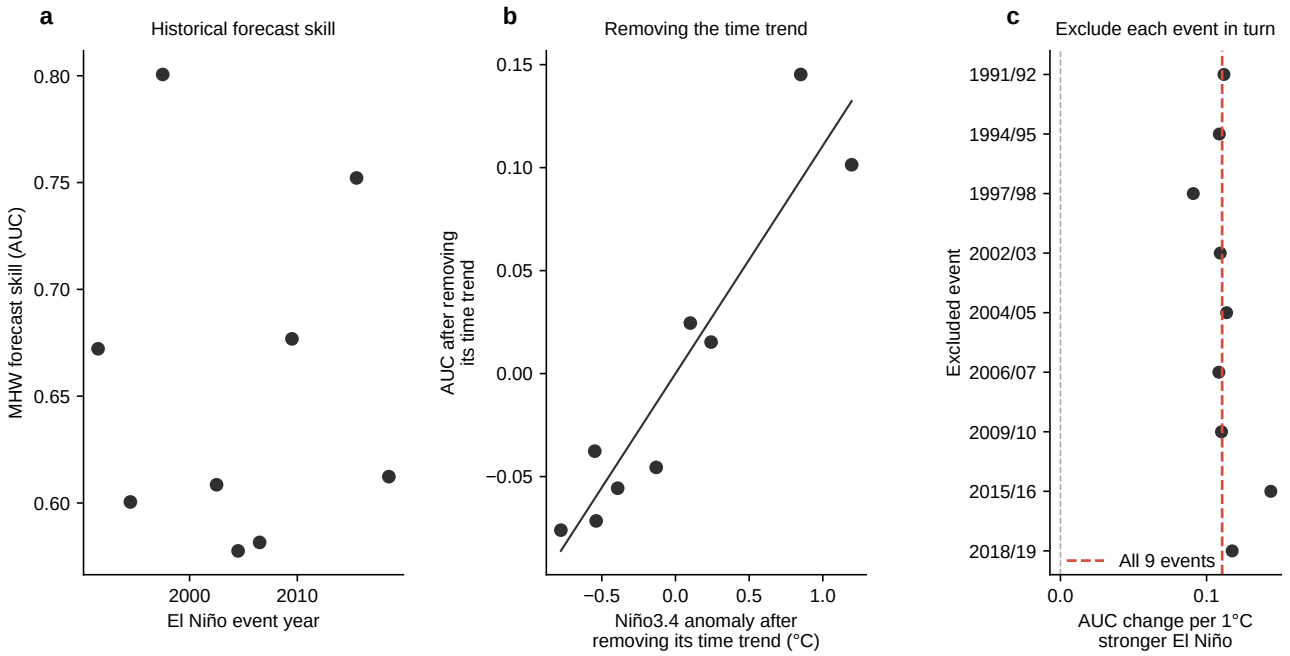
Supplementary Fig. 3 | Sensitivity of the ENSO–skill relationship to event selection



a, Mean AUC versus midpoint year for the nine earlier El Niño events. b, Niño3.4 amplitude and AUC after removing their respective linear relationships with event year. c, The Niño3.4 coefficient in AUC = intercept + amplitude coefficient × Niño3.4 + year coefficient × event year, refitted after omitting each named event. Each point uses eight events; the red dashed line is the coefficient from all nine events. The 2023–2024 event is excluded throughout. These nine omission estimates describe sensitivity to event selection; their spread is not a confidence interval and they are not nine independent experiments. Adjustment for event year assesses linear temporal confounding only.

Nine historical events; deterministic ordinary-least-squares fits. Event amplitudes and AUCs are identical to main Fig. 1a.